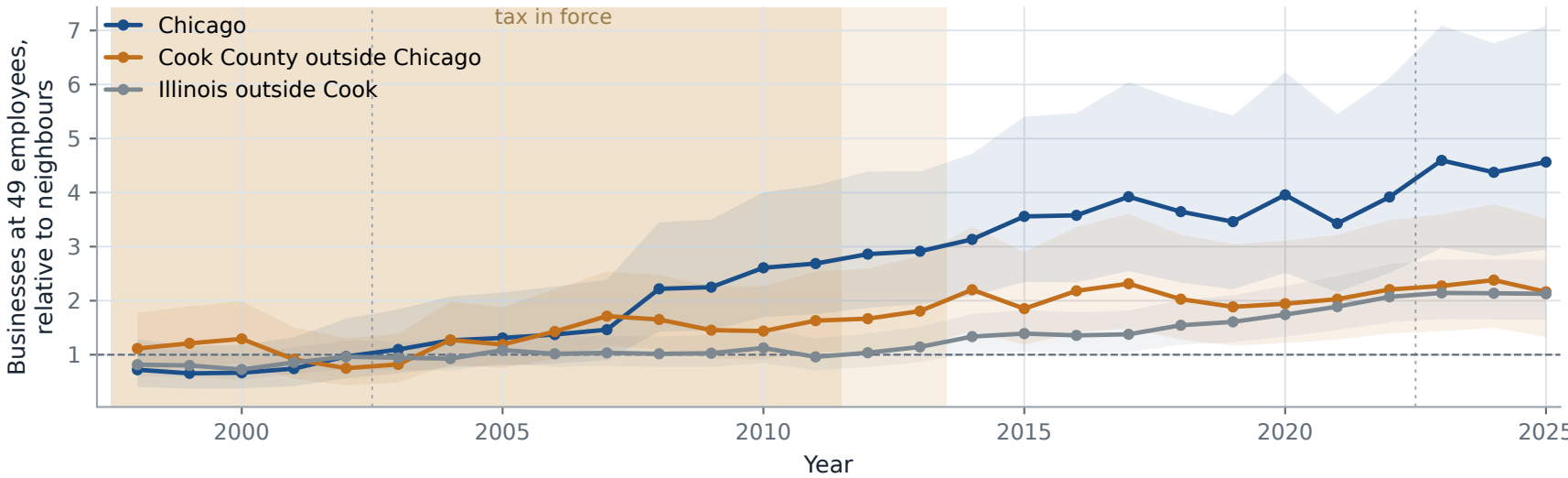


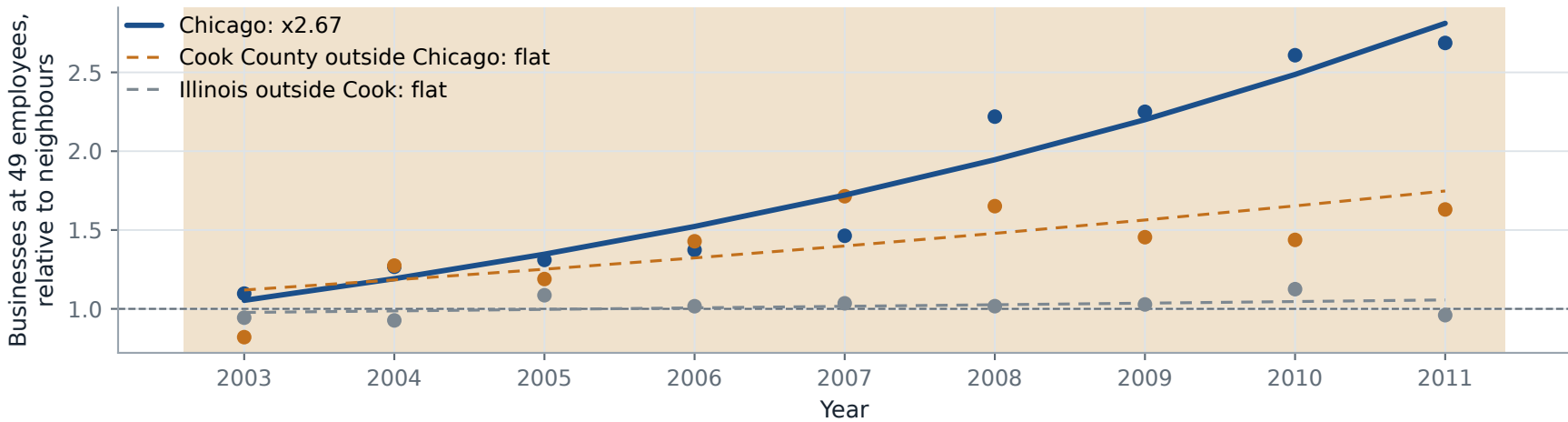
No evidence the head tax caused clustering below 50 employees

Businesses reporting exactly 49, the last size below the threshold, Chicago and two areas that never levied the tax, 1998-2025

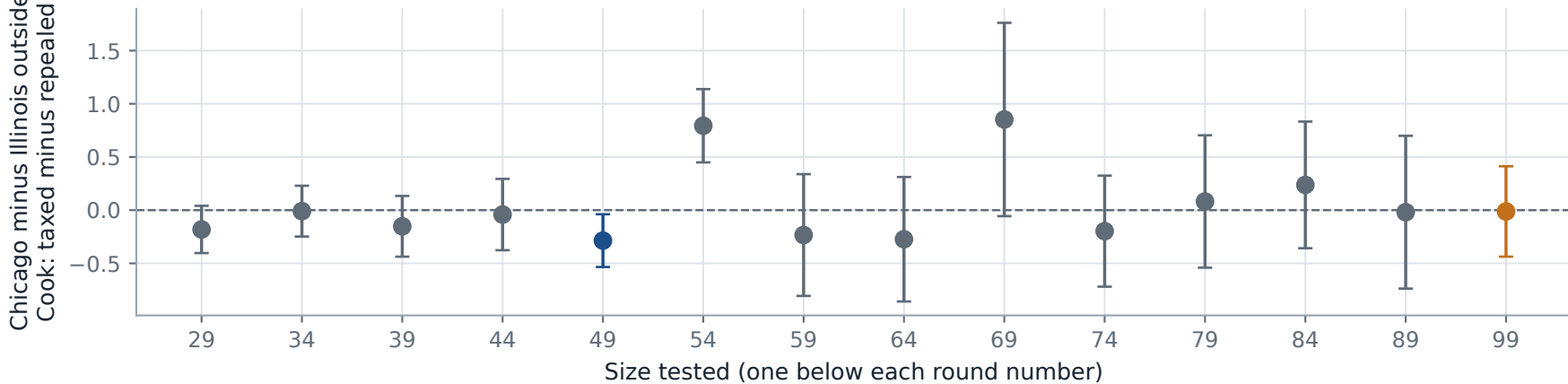
Clustering at 49 is lowest while the tax applies and highest after it is repealed



2003 to 2011: the rate, the threshold and the snapshot month were all unchanged



The tax test at 49, and the identical test at sizes where there is no tax to find



Top: clustering at 49 in Chicago runs 0.72 in 1998 and 4.57 in 2025, rising through the 2012 phase-out and the 2014 repeal, and rising in both comparison areas, neither of which levied the tax.

Middle: the rate stood at \$4 and the threshold at 50 from 1995 to 2011, and the snapshot month is July throughout 2003-2011. Chicago's clustering nonetheless rises by a factor of 2.67 across that window (+0.123 per year, +0.062 to +0.183). A series that moves this much while the policy is frozen is not tracking the policy. Over the same window Cook County outside Chicago by 1.56 and Illinois outside Cook by 1.08, neither distinguishable from zero, so the movement is largest in Chicago and its cause is unknown. It is not unique to 49 even within Chicago: size 39 rises significantly over the same nine years, and no employment threshold sits at that size.

Bottom: taxed years (2003-2011) against repealed years (2014-2022) in log points; negative means Chicago's gap over the comparison area widened after repeal. The estimate at 49 is -0.29 (-0.53 to -0.04) against Illinois outside Cook and -0.33 against Cook outside Chicago. Nominal intervals overstate precision: the identical test at 12 sizes with no tax returns -0.27 to +0.85, 1 of them nominally significant at 5%. Judged against that spread, 2 of 12 placebos are at least as large as the estimate at 49 ($p = 0.17$). Pooling 46-49 rather than testing 49 alone gives -0.18 (-0.35 to -0.01). The detection floor is correspondingly high: only an effect of roughly 0.9 log points, enough to nearly triple the clustering at 49, would have registered. No positive control exists in this data, so the estimator's sensitivity to genuine avoidance is not demonstrated.

Clustering is the count of businesses one below a round number divided by the average count at four, three and two below and two, three and four above, excluding the round number and the size just above it. Single-location businesses with self-reported headcounts; 1997 excluded for lack of a verification flag. Dotted lines mark vendor coding changes. Source: Data Axle (Infogroup) historical business files, 1998-2025.